Name	Chong Yew Sze
Student ID	35400552
Studio Session	Tutorial 05 (Friday 2-4pm)

Part A: URL of the Visualisation

https://yewsze.github.io/Data-Visualisation-2/

Part B: Domain Description

Describe your domain chosen for DV II

The visualisation focuses on Outdoor Recreation Participation in Australia. It focus on how adults take part in physical activity outdoors, who takes part, where it happens, and how it connects to the places and industries that depend on it. The page begins with national participation, narrows to the people and activities involved, and then moves to the places where outdoor activity takes place. It closes with the Northern Territory's national parks and the wider tourism economy.

Describe the Why and Who aspects of your DV II

Why:

Many Australians assume that a country with beaches, bushland and parks is naturally active. Yet participation is uneven across age groups, employment groups and states, and the reasons people take part vary. The visualisation addresses this need by showing who is outdoors, what they do and where they do it. Its aim is to show that outdoors is not just a place for people but part of how they live. The page communicates this through a story rather than acting as an expert exploration tool.

Who:

The audience is the average Australian adult, with no statistical background. Text lines are short, terms such as "participation rate" and "organised activity" are explained in plain language, and every chart carries a short annotation stating its main finding. It is also relevant to anyone curious about how Australians spend their leisure time, including local residents and domestic visitors.

Describe the "What" aspect of your DV II

The data comes from the below sources:

- Australian Bureau of Statistics. (2023, December 1). How Australian generations spent their time on recreation and leisure. https://www.abs.gov.au/articles/how-australian-generations-spent-their-time-recreation-and-leisure
- Australian Sports Commission. (n.d.). AusPlay results. Clearinghouse. https://www.ausport.gov.au/clearinghouse/research/ausplay/results
- Director of National Parks. (2025). Director of National Parks annual report 2024-25. Department of Climate Change, Energy, the Environment and Water. https://www.dcceew.gov.au/sites/default/files/documents/dnp-annual-report-2024-25.pdf
- Department of Tourism and Hospitality. (n.d.). Park visitor data. Northern Territory Government. https://dth.nt.gov.au/parks-and-wildlife/parks-and-wildlife-statistics-and-research/park-visitor-data"
- Australian Bureau of Statistics. (2025, December 17). Australian National Accounts: Tourism Satellite Account, 2024-25. https://www.abs.gov.au/statistics/economy/national-accounts/tourism-satellite-account/latest-release
- Zhao, S. (n.d.). Australian spatial data. GitHub. https://github.com/sherwinzxw/australian-spatial-data

Describe the "How" aspect of your DV II

Chart 1: Six in Seven Aussies Are Active (Pictogram)

- Why:
  To open the story with a positive headline: most adults take part in physical activity.
- How:
  Rows of person icons, with active adults as filled icons in a strong colour and the rest in grey. Counting icons is easier than reading a percentage, so the "6 in 7" message lands without axes or statistics.

Chart 2: Share of adults by activity level, Australia, 2025 (Waffle Chart)

- Why:
  To follow the headline with the exact share of adults who are active, so the viewer sees the precise ratio behind it.
- How:
  A 10 x 10 grid of 100 squares, where each square is 1% of adults. Active and not-active squares use two distinct colours, so the split reads at a glance. Equal-sized squares let readers judge the proportion without reading a number.

Chart 3: Most popular activities among adults, ranked by participation, 2025 (Ranked Bar Chart)

- Why:
  To show which activities draw the most adults and where outdoor activities sit among them.
- How:
  Horizontal bars sorted from highest to lowest participation, with long activity names readable on the left. A category filter lets users narrow the list to core outdoor activities. Bar length on a common baseline is the most accurate channel for comparing rates.

Chart 4: Type of activity by age group, share of adults, 2025 (Normalised Stacked Bar Chart)

- Why:
  To show whether adults are active through sport, non-sport activity, both, or neither, and how these changes with age.
- How:
  One bar per age group, each scaled to 100% and split into the four groups. Normalising makes age groups of different sizes directly comparable, and the "neither" segment shows how many adults are not active at all.

Chart 5: Population Pyramid of Adults Who Are Active at Least Once a Year, 2025 (Population Pyramid)

- Why: To compare male and female participation across age groups at a glance.
- How: Two back-to-back horizontal bars per age band, one for men and one for women, showing the share of adults active at least once a year. The mirrored layout makes gender gaps easy to see within each age band and across age.

Chart 6: Participation in core outdoor activities by age group, 2025 (Heatmap)

- Why: To show where each core outdoor activity peaks across age groups.
- How: A grid with activities as rows and age groups as columns, with cell colour (a sequential scale) showing participation. Colour makes it easy to scan a large grid for patterns, and a tooltip gives the exact value.

Chart 7: Main reasons for taking part, by age group, 2025 (Heatmap)

- Why: To show how motivations, such as fitness, enjoyment or socialising, shift across age groups.
- How: The same grid layout as Chart 6 but with motivations as rows. Reusing the layout means readers already know how to read it, and a shared colour scale helps them spot which reasons matter most to each age group.

Chart 8: How activity is organised, by age group, 2025 (Alluvial Chart)

- Why:
  To show how adults of different ages take part in organised and informal activity, and which kinds of organisations they use.
- How:
  Age groups on the left flow into how the activity is organised, with the width of each band showing the number of adults. The flows let viewers follow one age group through to its organisation type, which a bar chart can't show in one view.

Chart 9: Rank of places where adults are active, by age group, 2025 (Bump Chart)

- Why:
  To bridge from people to places by showing that public space is where much of the activity happens.
- How:
  Lines trace the ranking of all nine places for activity across age groups. “Public space” is highlighted in one colour and all other places are grey, so the viewer follows one line while the rest stay as context. Rank is shown rather than percentage so changes in order stand out.

Chart 10: Adults Activeness by State and Territory, 2025 (Multiple Choropleth Map)

- Why:
  To show how adult participation differs across states and territories.
- How:
  Three maps of the same geography, one each for at least once a year, at least once a week and three or more times a week, shaded by participation rate. Repeating the same map lets the viewer compare frequency levels directly, and a sequential colour scale suits rates.

Chart 11: National Parks and Reserves in the Northern Territory, by region (Dot Map)

- Why:
  To move the story from the national view to specific places and show where NT parks are located.
- How:
  Park locations as points over a pale NT basemap, coloured by NT region, with neighbouring states as muted context. Points suit locations because they keep each park distinct without implying area.

Chart 12: Most Visited Parks and Reserves in the Northern Territory, 2024-25 (Proportional Symbol Map)

- Why:
  To show which NT government parks are the busiest, so the viewer sees that visitors concentrate in a few places.
- How:
  Circles placed at each park and sized by visitor numbers, covering the top 11 parks. Area encodes the count, so the big destinations stand out immediately and the map doubles as a ranking.

Chart 13: Activity level of adults by employment status, 2025 (Mosaic Plot)

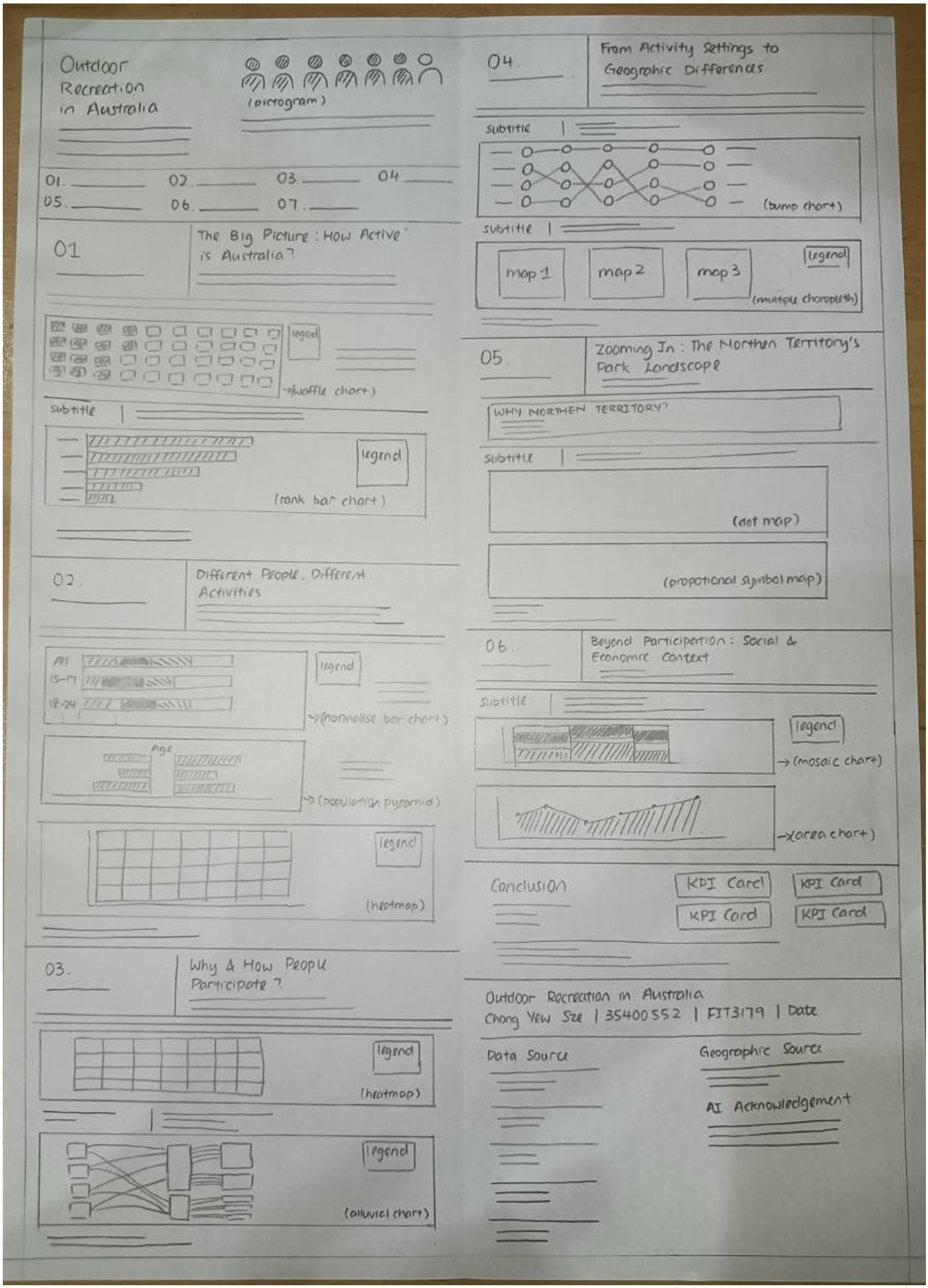
- Why:
  To show how often adults are active, and how employment status changes the picture.
- How:
  Columns for each activeness level (at least once a year and so on), with column height showing the percentage of adults and each column split by employment status. This shows both how common each frequency is and who makes it up, in one view.

Chart 14: Output of sport and recreation services in tourism, Australia (Area Chart)

- Why:
  To close the story by showing that sport and recreation support tourism output, not just personal well-being.
- How:
  An area chart of sport and recreation services output over time, with the 2020–21 pandemic drop annotated. The filled area shows the size of the industry as well as its trend, which supports the conclusion that the outdoors is part of how Australians live.

Part C: Sketch

https://github.com/yewsze/Data-Visualisation-2/blob/main/sketch%202.pdf

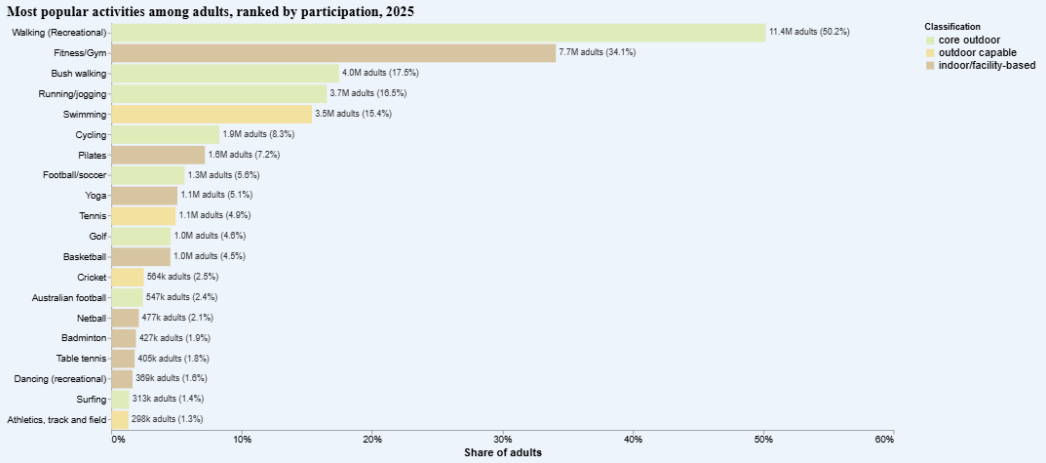

Outdoor Recreation in Australia
(pictogram)
01.
02.
03.
04
05.
06.
07.
01
The Big Picture : How Active is Australia?
legend
(waffle chart)
subtitle
legend
(rank bar chart)
02.
Different People, Different Activities
All
15-17
18-24
legend
(normalise bar chart)
Age
(population pyramid)
legend
(heatmap)
03.
Why & How People Participate ?
legend
(heatmap)
legend
(alluvial chart)
04.
From Activity Settings to Geographic Differences
subtitle
(bump chart)
subtitle
map 1
map 2
map 3
legend
(multiple choropleth)
05.
Zooming In : The Northen Territory's Park Landscape
WHY NORTHEN TERRITORY?
subtitle
(dot map)
(proportional symbol map)
06.
Beyond Participation : Social & Economic Context
subtitle
legend
(mosaic chart)
(area chart)
Conclusion
KPI Card
KPI Card
KPI Card
KPI Card
Outdoor Recreation in Australia
Chong Yew Sze | 35400552 | FIT3179 | Date
Data Source
Geographic Source
AI Acknowledgement

Part D: Full-size screenshot of your DV II



Activity ranking

Activities range from walking to organised sport. Ranking them shows where to look closer.



Filter by classification: Select All

National rankings can **hide differences between groups**. Next, we look at age and sex.

02

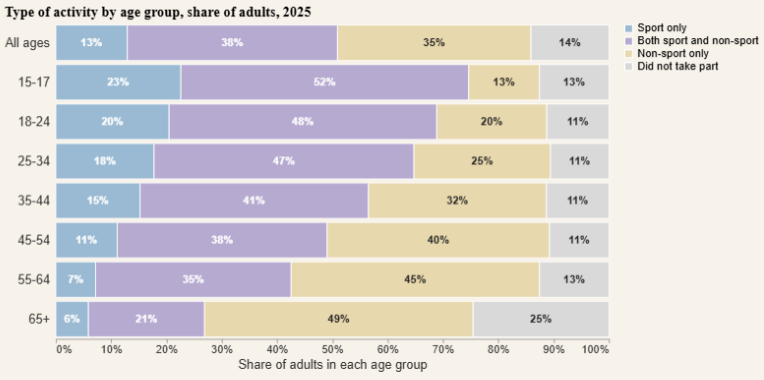
LOOK CLOSER AT CHOICES

Different People, Different Activities

Participation is **not uniform across Australia**. We look at who takes part, and what different groups choose.

Sport and Non-Sport by Age Group

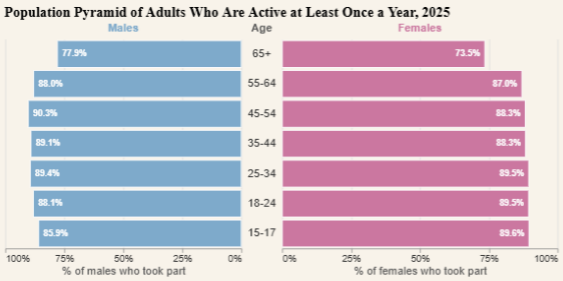
Do younger and older Australians favour different types of activity? This compares the mix of sport and non-sport.



Broad types **hide detail**. Next, we look at age and sex more closely.

Adults Active by Age and Gender

This looks closer at who takes part. It can show patterns that an overall rate hides.

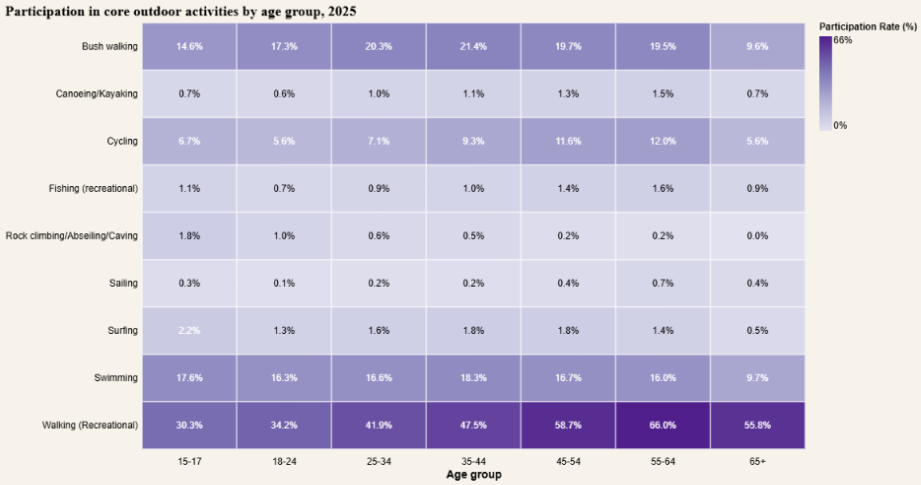


Bars show the share of each age and sex group, **not head counts**.

This shows who takes part, not what they do. Next, we **compare activities by age**.

Core Outdoor Activities by Age Group

Age groups may favour different activities. This shows which activities are more common in each group.



We now know who takes part and what they do. Next, we ask **why, and how activity is organised**.

03

LOOK BENEATH THE NUMBERS

Why and How Do People Participate?

Popular activities and demographics are **only part of the story**. Motivations and organisation add context.

Motivations by Age Group

Why do people take part, and does it change with age? This compares the reasons people give.

Main reasons for taking part, by age group, 2025

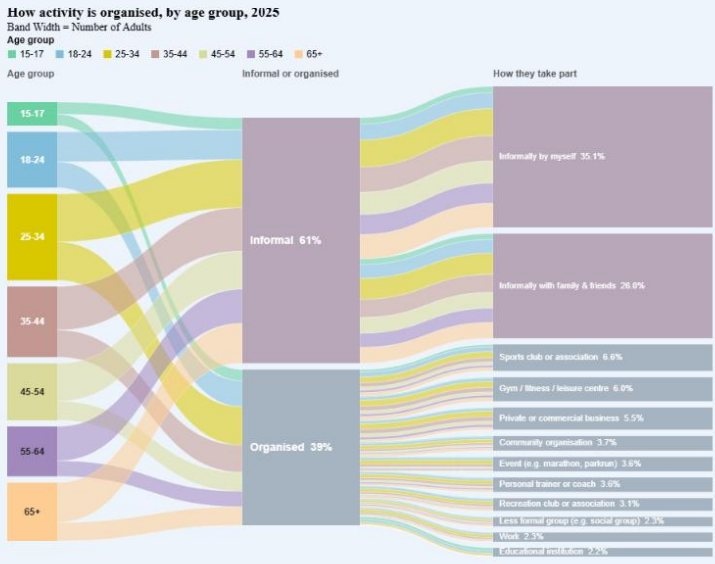
	All adults	15-17	18-24	25-34	35-44	45-54	55-64	65+
Physical health or fitness	67%	62%	62%	63%	63%	67%	71%	74%
Fun / enjoyment	59%	72%	65%	62%	60%	57%	58%	50%
Weight control or toning	53%	39%	50%	55%	55%	58%	60%	46%
To be outdoors / enjoy nature	51%	47%	47%	49%	50%	52%	58%	53%
Social reasons	44%	57%	55%	49%	42%	39%	39%	40%
Mental health or therapy	38%	27%	37%	42%	41%	41%	39%	30%
Sense of achievement	31%	38%	37%	33%	29%	27%	29%	28%
Hobby	29%	53%	47%	41%	31%	21%	17%	12%
Training	21%	35%	31%	27%	22%	19%	15%	8%
Walk the dog	20%	13%	16%	19%	19%	23%	26%	19%
Learn a new skill	15%	33%	26%	22%	14%	9%	8%	6%
Performance or competition	14%	34%	24%	17%	13%	10%	9%	9%

Share of participants: 80%, 60%, 40%, 20%, 0%

Motivations explain why people take part, not how activity is arranged. Next, we look at **who organises it**.

Organisers of Activity by Age Group

Activity can be organised or informal. Organised activity is run by clubs, commercial providers and others.



Bands show shares of participation, **not people moving between groups**.

People differ in their reasons and arrangements. Next, we move **from people to place**.

04

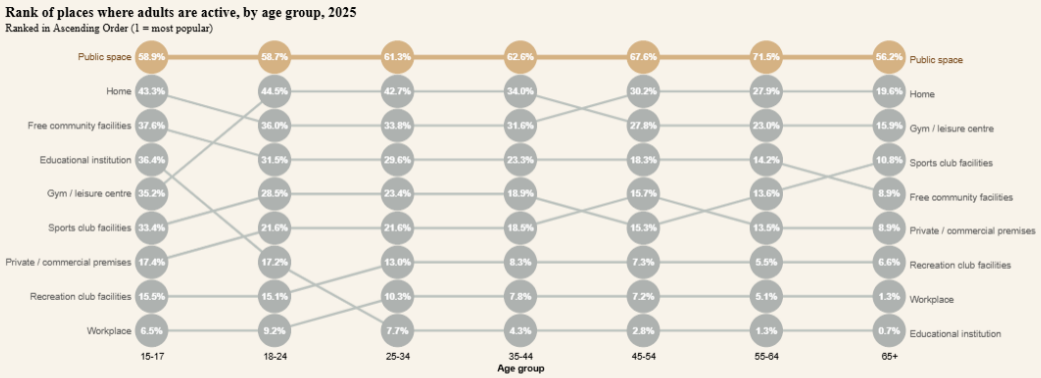
MOVE FROM PEOPLE TO PLACE

From Activity Settings to Geographic Differences

Where activity happens matters. We look at reported settings, then at **differences between states and territories.**

Where activities take place

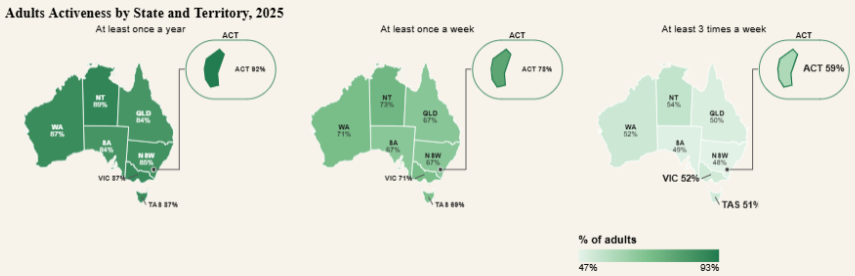
Recreation can happen in many settings. This compares how reported settings rank against each other.



Settings show where activity happens. Next, we ask whether participation **varies by location**.

Participation by State and Territory

Settings are one view of place. This map compares adult participation across states and territories, from occasional to regular.



State rates do not show individual parks or how often they are visited. So we **zoom into the Northern Territory**.

05

A CLOSER LOOK AT ONE TERRITORY

Zooming In: The Northern Territory's Park Landscape

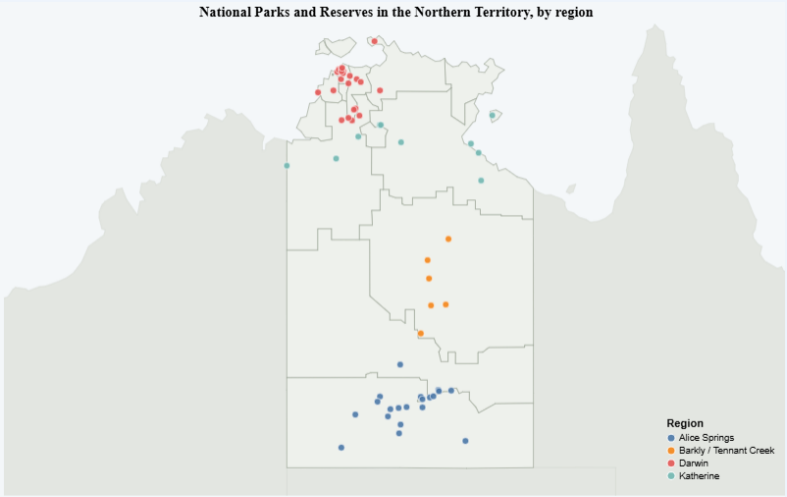
State rates cannot show **individual destinations**. The NT lets us look at parks run by the NT Government and the Commonwealth.

WHY THE NORTHERN TERRITORY?

The NT is one of the higher-ranking states and territories and is geographically extensive. Its parks also report visitor numbers. This is a case-study choice, **not evidence of better access to parks**.

Where Northern Territory Parks Are Located

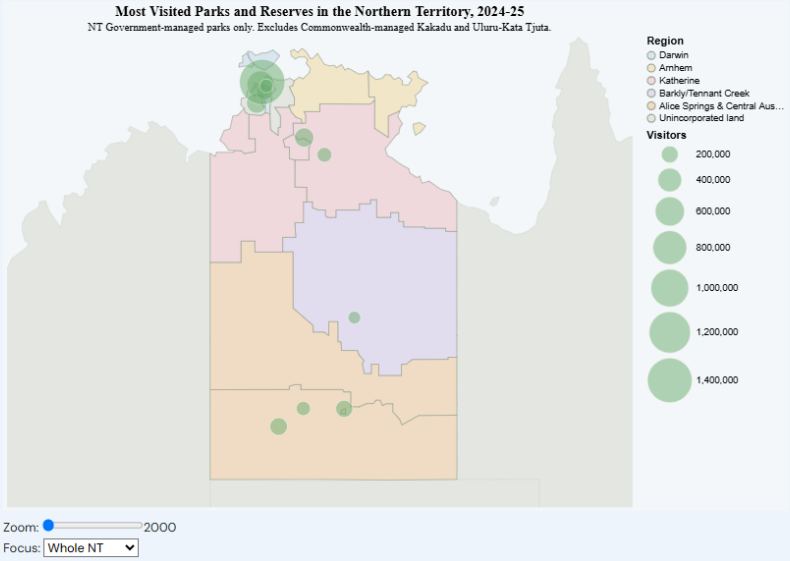
Where are the parks, and who manages them? The map separates NT Government sites from Commonwealth-managed Kakadu and Uluru-Kata Tjuta.



Location alone **does not show visits**. Next, we compare parks with comparable visitor estimates.

Top Visited Northern Territory Government Parks

Which NT Government parks have the highest estimated visits? Symbol size shows reported visitor numbers.



Visitor estimates use the NT Government's reporting period. Only parks with comparable data are shown, so this is **not a ranking of every NT park**. Kakadu and Uluru-Kata Tjuta are not included.

Park visits and participation **measure different things**. Next, we return to wider social and economic context.

06

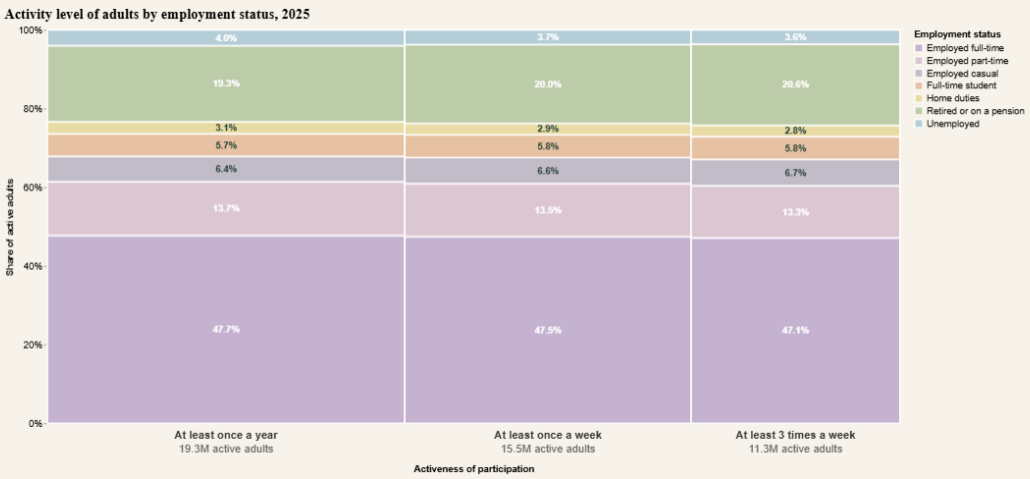
STEP BACK FOR PERSPECTIVE

Beyond Participation: Social and Economic Context

We end with **two wider views**: how participation relates to employment, and how tourism adds economic context.

Employment and Activity Level

Work shapes schedules and opportunities. This shows how activeness differs across employment groups.



This shows an association, **not a cause**.

Employment is one view of who takes part. Next, we **turn to the economy**.

Sport and Recreational Tourism Output

Beyond participation and park visits, tourism output gives an economic view. This shows how reported output changes over time.

Output of sport and recreation services in tourism, Australia



Output is an economic measure, **not a count of people**.

The Bigger Picture

No single chart explains outdoor recreation. Together, they show **how it fits into Australian life**.

From a national overview to a Northern Territory case study

Walking leads participation across selected outdoor activities.

Age, gender and geography shape these experiences.

Nature motivates participation and connects people with landscapes.

Outdoor recreation also contributes to **tourism and community activity**.

The outdoors is more than a destination.
It is part of how Australia lives.

Outdoor Recreation in Australia

Chong Yew Sze | 35400552 | FIT3179 | 26 October 2026

DATA SOURCES

Australian Bureau of Statistics. (2023, December 1). How Australian generations spent their time on recreation and leisure.

Time use and recreational activity patterns across Australian generations.

Australian Sports Commission. (n.d.). AusPlay results. Clearinghouse.

National and regional participation data on sports and physical activities in Australia.

Director of National Parks. (2025). Director of National Parks annual report 2024–25. Department of Climate Change, Energy, the Environment and Water.

Annual report covering management, performance, and conservation activities across Commonwealth reserves.

Department of Tourism and Hospitality. (n.d.). Park visitor data. Northern Territory Government.

Visitor statistics and research for parks and wildlife reserves in the Northern Territory.

Australian Bureau of Statistics. (2025, December 17). Australian National Accounts: Tourism Satellite Account, 2024–25.

Economic contribution, gross domestic product (GDP), consumption, and employment data for Australia's tourism industry.

GEOGRAPHIC SOURCES

Zhao, S. (n.d.). Australian spatial data. GitHub.

Simplified TopoJSON boundary data encoding Australian administrative regions.

AI ACKNOWLEDGEMENT

ClaudeAI was used to refine the storyflow, ensuring a coherent narrative throughout the report. Check grammar, assist with html and CSS styling and webpage layout, and debug issues.

Reading the data: Each chart has its own source and time period. Sport participation, park visits and tourism output measure different things, so compare them with care.